



PSYCHOLOGY

Stage 3

WACE Examination 2014

Marking Key

Marking keys are an explicit statement about what the examiner expects of candidates when they respond to a question. They are essential to fair assessment because their proper construction underpins reliability and validity.

Section One: Research methods

20% (34 Marks)

Question 1

(20 marks)

A psychologist was interested in studying the characteristics of young people who do volunteer work.

- (a) Name **two** non-experimental (descriptive) research methods that the psychologist could use. (2 marks)

Description	Marks
Any correct responses acceptable for one mark each. Examples of correct responses include:	
• Case study • Survey/questionnaire • Correlational study • Archival research • Observational study/observation	0–2
Total	2

- (b) Name **two** qualitative methods that the psychologist could use. (2 marks)

Description	Marks
Any correct response to be accepted for one mark each. Examples of correct responses include:	
• Interview (fixed response / open ended) • Focus group • (Open ended) questionnaire • Survey • Observational study/observation • Archival research • Checklist	0–2
Total	2

- (c) The psychologist designs a subjective quantitative measure to collect information about the personality characteristics of volunteers, using a Likert scale.

- (i) Name each of the **five** points on the line below to show a Likert scale. (1 mark)

Description	Marks
For 1 mark, all five points on the scale must be labelled correctly with a name, not a number. The 'Neutral' point may also be named 'Undecided' / 'Don't know' or other equivalent neutral response Correct response:	
Strongly disagree Disagree Neutral Agree Strongly agree	1
OR	
Strongly agree Agree Neutral Disagree Strongly disagree	

Note: Numbers (1, 2, 3, 4, 5) are not acceptable responses.	
Total	1

- (ii) Describe **one** advantage of using a Likert scale compared with a checklist.
(2 marks)

Description	Marks
Any correct answer acceptable for 2 marks. Two marks for a correct response that explicitly compares a Likert scale with a checklist by <i>referring to characteristics of both</i>. Example of 2 mark response: - A Likert scale allows the person to indicate the strength of agreement / how much they agree with a certain statement, whereas a checklist only provides Yes/No information. One mark for a correct response that describes an advantage of a Likert scale or a disadvantage of a checklist <i>but does not explicitly compare the two</i>. Examples of 1 mark responses include: - A checklist does not provide as much detailed information. - A checklist does not provide information about the strength of rating / how much a person agrees with a statement / item. - A Likert scale shows strength of a rating Note: A Likert scale produces data that can be analysed statistically / 'quantitative data' is not acceptable for one mark.	0–2
Total	2

The psychologist was interested to find out whether there is a relationship between the number of hours spent volunteering and self-esteem in young people.

- (d) Is this an experimental or correlational study? Explain the reason for your response.
(2 marks)

Description	Marks
Correlational study (1 mark) Explanation – any of the following (or equivalent) accepted for 1 mark. - the study does not have independent and dependent variables - the research is looking at the relationship between the variables - the research is not manipulating any variables	0–2
Total	2

The psychologist collected data on the self-esteem of ten young people who were currently involved in volunteer work in their community, using the Likert scale.

The Likert scale produced scores for each person ranging from 10 to 50. Higher scores indicated higher levels of self-esteem.

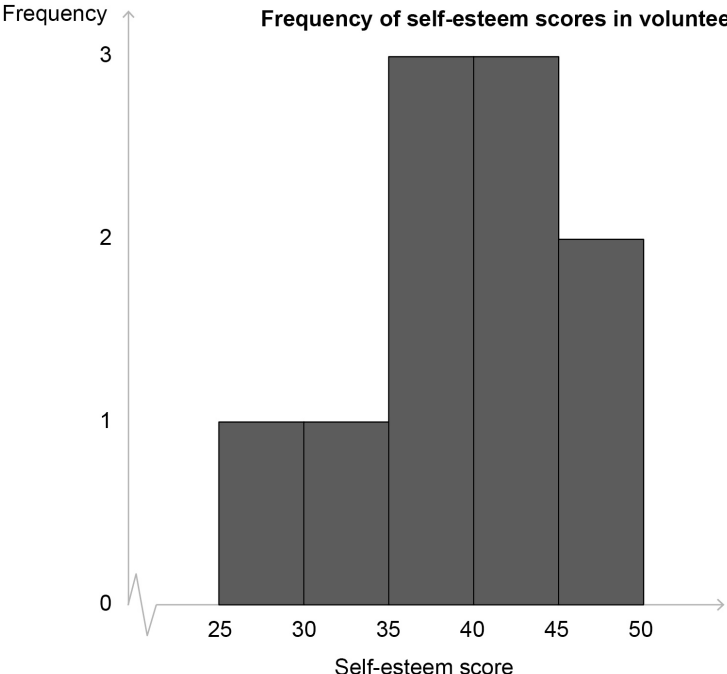
The self-esteem scores for the 10 young people were:

30, 32, 38, 40, 45, 43, 46, 38, 45, 46

- (e) (i) Complete the frequency table below. (1 mark)

Description		Marks
Every number must be correct for one mark. No other responses are accepted.		0–1
Self-esteem score	Number of people	
11–15	0	
16–20	0	
21–25	0	
26–30	1	
31–35	1	
36–40	3	
41–45	3	
46–50	2	
Total	10	
Total		1

- (ii) The psychologist is writing a report on the results of the research and needs to include a graph of the frequency of self-esteem scores. Graph the results from the frequency table in part (e) (i) on the axes below. (6 marks)

Description	Marks
One mark to be allocated to each of the following points: • Horizontal axis correctly labelled (self-esteem score) (1 mark) • Units on the horizontal axis to be marked correctly (1 mark) • Vertical axis to be correctly labelled (number of students or frequency) (1 mark) • Units on the vertical axis to be marked correctly (0–3, or may be higher) (1 mark) • Use of a column / bar graph (not a line graph) (1 mark) • Correct graphing of all data from the table (1 mark) Columns should be touching as this is a continuous scale but no penalty if the columns have a space between them. 	0–6
Total	6

The psychologist then collected data from a sample of ten young people who had never done any volunteer work.

The psychologist calculated the mean self-esteem score for the volunteer group and the non-volunteer group.

	Mean self-esteem score
Volunteer group	40.3
Non-volunteer group	38.5

A statistical test of the difference between the two means showed $p > 0.05$.

- (f) (i) State whether the difference between the two means was statistically significant. (1 mark)

Description	Marks
No (the difference between the means is not statistically significant)	0–1
Total	1

- (ii) Describe the meaning of 'statistically significant'. (2 marks)

Description	Marks
Any correct answer acceptable for up to 2 marks. Example of a 2 mark response: - Whether the difference between two scores is likely to be a true difference or due to chance factors / error Examples of 1 mark responses: - Whether the result is due to chance / error - Whether the difference is real	0–2
Total	2

- (iii) Outline the conclusion that the researcher should draw from the statistical results. (1 mark)

Description	Marks
Any response that indicates that there was no significant difference between the scores of the two groups accepted for one mark. Examples of correct responses include: - The young people who did volunteer work did not have higher self-esteem scores than the young people who did not do volunteer work. - The young people in the two groups were not significantly different in their levels of self-esteem. - The self-esteem scores of the young people who did volunteer work and did not do volunteer work were the same.	0–1
Total	1

Question 2**(14 marks)**

Professor Zheng of Smart University wanted to test whether a vitamin supplement helped to improve students' grades. To test her idea, she told her class she was giving them all a supplement that would make them smarter. She told them they all had to participate.

Professor Zheng divided the class into two equal groups. Group A received the vitamin supplement and Group B received an inactive substance (sugar pill).

Over the two week trial, Professor Zheng spent a lot of time with the students gathering data. She was excited to find the results showed that all students' grades had improved. She concluded that the vitamin supplement had worked.

- (a) (i) Name **three** rights people have as participants in research. (3 marks)

Description	Marks
Any correct response accepted for one mark each. Examples of correct responses include, but are not limited to: • Voluntary participation / the right to choose to participate • Privacy • Anonymity • Confidentiality • Right to withdraw / withdrawal Note: 'Informed consent' is not a right and not accepted for one mark.	0–3
Total	3

- (ii) Outline **two** pieces of information about a research study that a researcher must provide to participants so that they can give informed consent. (2 marks)

Description	Marks
Any correct response acceptable for one mark. Examples of correct responses: • What the research involves / what they will be required to do • Information about any risks of participating • Information about any benefits of participating Note: Responses such as 'right to withdraw', 'confidentiality' and so on are not accepted as these are not pieces of information about a research study.	0–2
Total	2

- (b) (i) Identify term that refers to a substance with no known medical effects used as a control in an experiment. (1 mark)

Description	Marks
Placebo	0–1
Total	1

- (ii) Explain why Professor Zheng gave one group a substance with no known medical effects. (1 marks)

Description	Marks
Any correct explanation of the function of a placebo acceptable for one mark. Correct response does not need to include the term 'placebo'.	0–1
Examples of correct responses:	
- A placebo was used to test if the effect was due to the expected benefits of taking a tablet rather than taking the vitamin. - To control for an expectancy effect	
Total	1

- (c) Explain what Professor Zheng should do after the experiment has ended to make sure the research has been conducted ethically. (1 mark)

Description	Marks
Debrief the participants (by explaining to them why the deception was used)	0–1
Total	1

- (d) Outline how Professor Zheng may have affected the results of the experiment. (1 mark)

Description	Marks
Any response that refers to an example of experimenter bias or experimenter effect accepted for one mark.	0–1
Examples of correct responses:	
- The (positive) results might have been because Professor Zheng spent a lot of time with the students collecting data for the study. - Professor Zheng might have marked students more leniently / given students higher marks because she wanted the experiment to work.	
Note: Responses such as 'experimenter bias' or 'experimenter effect' are not explanations and are not acceptable for one mark.	
Total	1

- (e) (i) Outline **three** ethical issues associated with this research. (3 marks)

Description	Marks
Any correct response accepted for 1 mark each. Examples of correct responses include: • No informed consent • The students were required to participate in the experiment • Deception / some students did not actually receive the supplement but thought they were • Relationship between the experimenter/professor and the participants/students • There may be benefits of the vitamin that placebo group have not had access to	0–3
Total	3

- (ii) Identify **two** sources of error in the design of this experiment. (2 marks)

Description	Marks
Any response that identifies a source of error in the research design accepted for 1 mark. Examples of correct responses: • There was no control group that did not receive any pill/substance at all • The researcher was the teacher (and could have been biased) / the researcher should have been independent • The experiment time frame was too short • Unequal groups (e.g. different age) • Hawthorne effect (all students were told the supplement would make them smarter) Note: Responses that refer to aspects that are not part of the research design e.g., low participant numbers / unknown gender distribution, culture and so on are not accepted for one mark.	0–2
Total	2

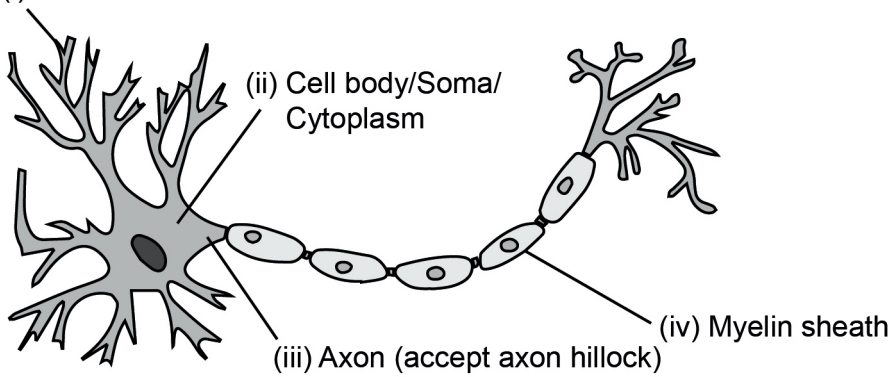
Section Two: Short answer

55% (90 Marks)

Question 3

(13 marks)

- (a) Identify the structure of a neuron shown by the labels in the illustration below. (4 marks)

Description	Marks
 (i) Dendrites (ii) Cell body/Soma/Cytoplasm (iii) Axon (accept axon hillock) (iv) Myelin sheath	0–4
Total	4

- (b) Describe the purpose of the structure labelled at (iv) in part (a). (1 mark)

Description	Marks
Any correct response accepted for one mark. Examples of correct responses: • Helps improve the conduction of nerve impulses along the axon • Increases the speed of conduction of electrical impulses / transmission of messages • Insulates the axon from chemical and physical stimuli that might interfere with the transmission of nerve impulses • Protects the axon	0–1
Total	1

- (c) Describe how neurotransmitters work in communicating messages between neurons at a synapse. (3 marks)

Description	Marks
Neurotransmitters carry a message from a sending neuron (1 mark) across a synapse/synaptic gap (1 mark) to a receptor site on a receiving neuron (1 mark)	0–3
Total	3

- (d) List **two** effects of each of the following neurotransmitters.

- (i) Dopamine (2 marks)
(ii) Serotonin (2 marks)

Description	Marks
Any correct responses acceptable for 1 mark each. Correct responses include, but are not limited to:	
(i) Dopamine regulates: • Emotional arousal • Pleasure / reward • Voluntary movement • Attention	0–2
(ii) Serotonin regulates: • Sleep • Emotional arousal • Aggression • Pain • Mood	0–2
Note: any specific examples of any of the above (or other correct) effects also acceptable.	
Total	4

- (e) Outline **one** way in which exercise can affect emotion. (1 mark)

Description	Marks
Any correct response acceptable for 1 mark.	
Examples of correct responses include: • Lowers blood pressure and blood pressure reaction to stress • Increases production of mood boosting neurotransmitters	0–1
Total	1

Question 4**(14 marks)**

Adam was at a party with friends from school. He was not enjoying himself and was thinking 'I want to go home'.

- (a) Identify **one** observable behaviour that Adam may be showing. (1 mark)

Description	Marks
Any reasonable response accepted for one mark. Examples of correct responses: • Sitting / standing by himself • Looking at his watch/phone • Yawning	0–1
Total	1

Suddenly Adam sees Lucy, with whom he lost contact many years ago. As she walks toward him, he remembers her name.

- (b) Identify what type of memory Adam used when he remembered Lucy's name. (1 mark)

Description	Marks
Long term memory/Declarative	0–1
Total	1

- (c) Adam and Lucy chat for a while and she says that they should stay in contact. Lucy tells Adam that her telephone number is 0425 456 789. Adam does not have anything to write the phone number down on, so he tries to remember it.

- (i) Describe briefly **one** problem that Adam might have in trying to remember Lucy's telephone number. Explain the reason why, referring to features of short-term memory. (2 marks)

Description	Marks
Examples of correct responses: • He might not be able to remember all of the numbers (1 mark) as short term memory capacity is limited (seven plus or minus two items) (1 mark) • If he has to remember the phone number for a long time before he can write it down he might forget the numbers (1 mark) because people can only hold information in short term memory for a short time (1 mark)	0–2
Total	2

- (c) (ii) Outline **two** strategies that Adam could use to assist him in remembering Lucy's telephone number. (2 marks)

Description	Marks
Any correct response acceptable for one mark each. Correct responses do not need to use correct technical terms/names. Examples of correct responses include, but are not limited to: • He could use chunking to remember the telephone number in groups of numbers / He could divide the number into parts and remember 0425 – 456 – 789 (1 mark) • He could remember the series of numbers 456789 (1 mark) • He could relate the number to something that he is familiar with/Develop a mnemonic (1 mark) • He could use rehearsal to say the number over and over to himself (1 mark)	0–2
Total	2

- (d) List **two** changes in cognition that can be experienced when a person has an altered state of consciousness. (2 marks)

Description	Marks
Any correct response acceptable for one mark each. Correct responses include, but are not limited to: • Shift of attention/awareness • Distorted perception/hallucination • Lack of memory for the time / amnesia • Changed sense of time • Distorted thinking • Increased suggestibility Note: Examples of changes in behaviour are not accepted for a mark. (e.g., loss of coordination, loss of control)	0–2
Total	2

- (e) Name **two** physiological responses that can indicate a change in a state of consciousness. Outline how each physiological response is different in sleep compared with waking consciousness. (4 marks)

Description	Marks
Correct responses (any correct brief description acceptable) Electrical activity of the brain (1 mark) - Electrical activity is highest during waking consciousness / brain waves are fast and small shape of brain waves changes during different stages of the sleep cycle (or any specific example of these different kinds of brain waves (1 mark) Heart rate (1 mark) - Heart rate is faster during waking consciousness / slower during sleep (1 mark) Body temperature (1 mark) - Body temperature is higher in waking consciousness / lower during sleep (1 mark) Galvanic skin response (1 mark) - Galvanic skin response is higher during waking consciousness and lower during sleep (1 mark) Blood pressure (1 mark) - Lower during sleep (1 mark)	0–4
Total	4

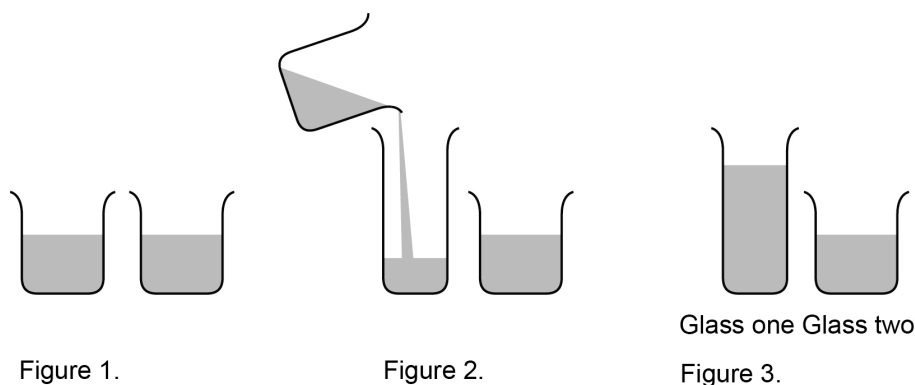
Question 5

(14 marks)

- (a) Name the stage of Piaget's theory of cognitive development that corresponds to each of the following characteristics: (4 marks)

Description	Marks
1 mark for each correct name. No other responses accepted.	
(i) Concrete operational (ii) Sensorimotor (iii) Formal operational / formal operations (iv) Pre-operational	0–4
Total	4

- (b) A psychologist has two glasses that are the same shape and size and are filled with equal amounts of water (see Figure 1). The psychologist pours the water from one glass into a tall thin glass (see Figure 2). The psychologist asks two children whether Glass one or Glass two has more water in it or whether both glasses have the same amount (see Figure 3).



Describe the response that each child would be **most** likely to give and explain the reason for the response.

- (i) Nikolas (age 4) (2 marks)

Description	Marks
Nikolas (age 4) will say that the glasses have different amounts of water in them. / Glass one has more / Glass two has less (1 mark)	
Examples of correct explanations: He might say Glass one has the most if he focuses on the width of the glass (1 mark) He might say Glass two has the most if he focuses on the height of the glass (1 mark).	0–2
Nikolas is in the pre-operational stage and does not understand that volume/mass does not change simply with a change of shape. (1 mark)	
Nikolas is in the pre-operational stage (1 mark)	
Nikolas can not conserve (volume/mass) (1 mark)	
Total	2

(ii) Hannah (age 10)

(2 marks)

Description	Marks
Hannah (age 10) will say that the glasses have the same amount. (1 mark)	0–2
Examples of correct explanations: Hannah is able to recognise that if no water was spilt or added, the two glasses will have the same amount of water, even though they might look different (1 mark)	
Hannah is able to consider both the height and the width of the glass at the same time to realise that the glasses still have the same amount of water in them (1 mark).	
Hannah is in the concrete operational stage and understands that volume/mass do not change simply with a change of shape. (1 mark)	
Hannah is in the concrete operational stage (1 mark)	
Hannah can conserve (volume/mass) (1 mark)	
Total	2

- (c) Dana is playing hide and seek with her brother Dion (aged three). Dion stands in the centre of the room and covers his eyes with his hands and says 'I am hiding'. Explain why Dion believes that he is 'hiding', referring to the relevant concept from Piaget's theory of cognitive development. (3 marks)

Description	Marks
Dion is displaying egocentrism (1 mark) which means that he has difficulty/is unable to consider other people's perspectives (1 mark). As he cannot see his sister, he believes that she cannot see him (1 mark).	0–3
Total	3

- (d) Name **one** researcher who has criticised the tasks used by Piaget and explain their criticism. (3 marks)

Description	Marks
Either of the following examples or other correct example, are acceptable Example 1 • Donaldson/Hughes (1 mark) • Criticised Piaget's tasks as being unfamiliar to children (1 mark) • When they are given more familiar tasks they are able to demonstrate higher abilities at a younger age (1 mark) Example 2 • Seigal (1) • Criticism related to language used in tasks (1 mark) • The experimenter asks questions that are too obvious or repetitive so children assume their first response was incorrect and change their response to please the experimenter (1 mark) Note: 1 mark for researcher 1 mark for main point being criticised 1 mark for an elaboration of the point being criticised	0–3
Total	3

Question 6

(10 marks)

(a) Define 'personality'.

(2 marks)

Description	Marks
Any correct definition of personality acceptable. 1 mark for characteristic ways of thinking / feeling / acting 1 mark for that make a person unique / an individual Example of 2 mark definition: "The characteristic ways of thinking, feeling and acting that make a person an individual" (Fletcher & Garton, 2007, p. 497)	0–2
Total	2

(b) Psychologists do not agree on whether personality is the result of genetics or environment or both. Outline what determines personality according to each of the following theories.

- (i) trait theory (1 mark)
- (ii) humanistic theory (1 mark)
- (iii) psychodynamic theory (1 mark)

Description	Marks
One mark for each correct response (i) Trait theory – traits are inherited / have a genetic/biological basis (but are also shaped by the environment / experiences) (ii) Humanistic theory – personality develops from experience / environment (ii) Psychodynamic theory – personality develops from experiences in childhood / environment	0–3
Total	3

(c) At age six, Anna has many friends and is very confident in speaking with adults. At age 17, Anna is the captain of her school debating team and a member of the student council.

- (i) Identify **one** personality trait that Anna has, according to Eysenck's trait theory. (1 mark)

Description	Marks
(High) Extraversion - Low neuroticism (must mention low) - Low psychoticism (must mention low)	0–1
Total	1

- (ii) Explain how trait theory would account for the consistency of Anna's personality over time. (1 mark)

Description	Marks
Any correct response acceptable for 1 mark. Examples of correct responses include: • Trait theories explain personality as being due to the presence of certain 'traits', which are enduring characteristics that make a person unique and that predispose them to behave in certain ways. • Traits are enduring characteristics • Because traits have a genetic basis they lead to consistent personality over time.	0–1
Total	1

- (d) At age 23, Anna gets her first full time job in a large company. At work, Anna eats her lunch at her desk rather than going out to lunch with her colleagues. However, on the weekends, she enjoys going out to lunch with her friends.

Explain, with reference to personality theory, why Anna's behaviour is not consistent across these situations. (1 mark)

Description	Marks
Any correct response acceptable for 1 mark. Examples of correct responses include: • Personality does not dictate how a person will act in every situation; it just helps to predict how a person tends to react. • Behaviour is usually the result of an interaction between the person and the situation (Mischel's 'person by situation' interaction)	0–1
Total	1

- (e) Outline **two** limitations of trait theories of personality. (2 marks)

Description	Marks
Any correct response acceptable for 1 mark each. Examples of correct responses include, but are not limited to: • Trait theories fail to take into consideration unconscious processes, beliefs and motives that may influence personality • There is no agreement on the number or types of traits that exist • Some people say that traits are not 'personality' but describe predispositions to behave in certain ways in certain situations • Relies on self-reports to measure personality traits	0–2
Total	2

Question 7**(7 marks)**

(a) Provide a psychological definition of 'attribution'.

(2 marks)

Description	Marks
Any correct response acceptable for up to two marks.	0–2
Response must make reference to making a decision/about or explaining a behaviour based on the perceived cause of behaviour.	
Examples of 2 mark responses include, but are not limited to: • “a decision about the basis for a person’s behaviour” Fletcher and Garton (2007) glossary • “decision about whether a person’s behaviour is due to something about them or to the situation they are in” (Fletcher & Garton, 2007, p. 54) • Making inferences about the causes behind the behaviour of others” (Fletcher & Garton, 2007, p. 442)	
Example of a 1 mark response: • What causes a person’s behaviour	
Total	2

(b) Sandie is waiting in a long line at the supermarket checkout to pay for a box of chocolates. Sandie is tapping her foot quickly and sighing loudly. An onlooker observes Sandie’s behaviour and thinks that she is a quick-tempered person.

(i) Explain, using attribution theory, why the onlooker is likely to think this. **(2 marks)**

Description	Marks
Any correct response acceptable for up to two marks.	0–2
Example of a 2 mark response: • Onlookers usually make an internal attribution for someone’s behaviour/perceive behaviour as being the result of internal factors (rather than considering external factors) so a person is likely to interpret Sandie’s behaviours as signs of impatience or quick-temperedness.	
Example of a 1 mark response: • She is making an internal attribution	
Total	2

- (ii) Describe **one** alternative attribution for Sandie's behaviour. (1 mark)

Description	Marks
Any reasonable response acceptable for 1 mark.	0–1
Correct responses include, but are not limited to:	
• Sandie is in a hurry • Sandie has someone waiting for her • Sandie needs to go to the toilet • Sandie is tired	
Total	
	1

- (c) Explain, using attribution theory, the difference between the attributions people tend to make for the behaviour of others compared with their own behaviour. (2 marks)

Description	Marks
Any correct response acceptable for up to 2 marks. For 2 marks, response must include explicit comparison of attributions made for own AND others' behaviour. Responses that refer to <i>either</i> own or others' behaviour should be awarded one mark only.	0–2
Examples of 2 mark responses:	
• When we explain the behaviour of others we tend to make internal attributions/attribute it to something about them as a person, whereas when we explain our own behaviour we tend to refer to the situation/ environment / external factors. This is known as the Fundamental Attribution Error. • We tend to explain the behaviour of others as due to some permanent/ongoing characteristic of theirs but when we explain our own behaviour we refer to temporary characteristics (e.g. feelings) or situational factors • We tend to explain the behaviour of others in more negative ways and our own behaviour in ways that put us in a more positive light (self-serving bias)	
Examples of 1 mark responses:	
• People tend to attribute their own behaviour to external factors • People tend to attribute other people's behaviour to internal factors • We tend to make more positive attributions for our own behaviour • We tend to make more negative attributions for other people's behaviour • People have a self-serving bias • Fundamental Attribution Error	
Total	2

Question 8

(12 marks)

- (a) Two groups of students want to use the same sports equipment at lunchtime at school. Name **three** different types of solutions that could be used to resolve this conflict and provide **one** example of each. (6 marks)

Description	Marks
1 mark for each correct name. Names must be as below. No alternative responses accepted. 1 mark for each correct example (must correspond to the name provided). Imposed solution (1 mark) Examples may include, but are not limited to (1 mark) - Teacher gives the sports equipment to one group of students - One group of students just takes the sports equipment Distributive solution (1 mark) Examples may include, but are not limited to: (1 mark) - Students might agree to take it in turns to use the sports equipment (e.g., for part of lunch time, or on different days/weeks) - Teacher might set up a booking system so that students can reserve the sports equipment ahead of time Integrative solution (1 mark) Example includes, but is not limited to (1 mark): - Students decide to play with the equipment together	0–6
Total	6

- (b) Namita is 13 years old and is starting at a coeducational high school. Describe, using the first **three** of Dunphy's stages of the changing structure of social groups during adolescence, how Namita's peer relationships are likely to change throughout high school. (6 marks)

Description	Marks
1 mark for correct name of each stage (exact term as below must be used). 1 mark for each correct description. Stage 1: Cliques (1 mark) Namita is likely to make friends with a group of girls (single sex group) (1 mark) Stage 2: Crowd (1 mark) Different cliques will start to interact with each other to form a crowd that has members of both genders (1 mark) Stage 3: Dyads/Couples (1 mark) Clique members of different genders may pair up in dyads / may start dating (1 mark)	0–6
Total	6

Question 9**(10 marks)**

- (a) Outline **one** reason why psychologists are interested in studying cultural influences on social and emotional development. (1 mark)

Description	Marks
Any correct response accepted for one mark. Examples of correct responses include, but are not necessarily limited to: - To better understand the factors that affect development - To understand similarities in development of people from different cultures - To understand why there are differences in development in people from different cultures Examples of 0 mark responses: - People do different things - People are different	0–1
Total	1

- (b) Cross-cultural research using Kohlberg's stages of moral development has concluded that the moral reasoning of people from Western cultures is at a higher level than people from non-Western cultures.

- (i) Explain whether this is an accurate conclusion to draw. (2 mark)

Description	Marks
Examples of correct responses: - No, this is not an accurate conclusion (1 mark) because Kohlberg's stages are based on western ideals of justice and individual rights (1 mark) - No, this is not an accurate conclusion (1 mark) because Kohlberg's theory does not take into account ideals/principles of eastern/collectivist cultures (1 mark)	0–2
Total	2

- (ii) Name the author(s) and describe **one** study that has challenged the conclusion that people from Western cultures reason at a higher level than people from non-Western cultures. (3 marks)

Description	Marks
Any correct response accepted for up to two marks. Date of research not required for a mark. Examples include, but are not limited to: • Santrock (1999) – Buddhist monks in Nepal did not rate high as they based their reasoning on compassion and prevention of suffering rather than justice • Shweder, Mahapatra, and Miller (1990/1987) – Indian and American subjects rated 39 scenarios in order of how serious the 'wrong' was. Results shows very different justifications are used when making moral judgements. Indians believed eating chicken and getting a haircut after father's death was far worse than beating a disobedient wife. • Snarey (1985) – collectivist happiness over individual happiness was noted in the Israeli Kibbutz community Note: 1 mark for author(s) 1 mark for participants 1 mark for finding	0–3
Total	3

- (c) Mr Eades told his doctor that he often heard his mother talking to him. Mr Eades's mother died two years ago. The doctor was very concerned and wondered if he should suggest that Mr Eades see a psychologist. Explain why the doctor should investigate Mr Eades's culture before making the suggestion. (1 marks)

Description	Marks
Examples of correct responses: • Hearing the voice of a loved-one who has passed away may be considered normal in some cultures but may be a sign of a psychological disorder/abnormality in other cultures. • What is considered normal in one culture might be considered abnormal in another culture.	0–1
Total	1

- (d) Mrs Costa is referred to a psychologist for cognitive-behavioural therapy because she has been feeling sad and worried. Explain **three** things that the psychologist could tell Mrs Costa about how cognitive-behavioural therapy might help her. (3 marks)

Description	Marks
One mark for each correct response. Correct responses include, but are not limited to: • Cognitive behavioural therapy (CBT) can help her identify and change negative thinking • CBT can help her motivation to do activities that she enjoys • CBT can help her learn coping strategies to better manage stress/problems e.g. relaxation strategies/ breathing exercises	0–3
Total	3

Question 10**(10 marks)**

- (a) Two small towns are on the edge of a large forest. People in one town want to cut down the trees in the forest and people in the other town want to protect the forest. For each town, describe **one** value that the people might hold. (2 marks)

Description	Marks
One mark for describing each town's values. Any correct response acceptable for one mark. Answers may include: • The town that wants to log may value economic/ industry growth/development (1) • The tree savers may value environment, nature, preservation, protecting resources, conservation (1)	0–2
Total	2

- (b) Different group values can create conflict. Provide **two** explanations why these differences can create conflict. Refer to concepts such as emotions, collectivist/individual cultures, perceptions and/or motives in your response. (4 marks)

Description	Marks
Two marks for each correct explanation. Correct answers may include, but are not limited to: • Values and emotions are linked – different values bring up emotions • May be collectivist culture reliant on each other rather than individualistic • Perception of motives can influence conflict (interpret rivals as malevolent)	0–4
Total	4

- (c) Identify **two** issues in society about which different social groups have different values. For each issue explain how the different values conflict. (4 marks)

Description	Marks
Any reasonable response accepted. 1 mark for issue 1 mark for explanation of the conflict of values Examples of correct responses may include, but are not limited to: • Abortion – conflict between the value of the right to life of the unborn child and the right of choice of the mother. • Euthanasia – conflict between the value of the right of the individual to choose to end pain and suffering and the value of the assisted death as immoral/illegal • Conserving wildlife – valuing financial success/development compared to conservation/safety • Same sex marriage – conflict between the traditional definition of marriage between a man and a woman compared to alternate views of marriage between two men or two women	0–4
Total	4

Section Three

25% (49 Marks)

Question 11

(26 marks)

Psychologists have researched and theorised about the influence that biological factors and the environment have on the development of intelligence and language.

Describe what twin and adoption studies reveal about the nature and nurture debate in relation to the development of intelligence (as measured by IQ). In your response you should

- describe examples of biological and environmental factors that can influence the development of intelligence
- describe how twin and adoption studies are carried out
- describe results that have been found
- describe conclusions that have been drawn from the results.

Discuss the conclusions that have been made about whether language is innate or learned with reference to **two** theories. In your response you should

- name the theorists and their theories.
- describe each theory and the evidence that is used to support it.

Description	Marks
Examples of biological and environmental factors that can influence the development of intelligence	3
Mark allocation: 1 mark per example to a maximum of 3 marks. Response must include at <i>least one example</i> of a biological factor and <i>at least one example</i> of an environmental factor. Examples Biological factors • Genetic predisposition/DNA/heredity • Teratogens, toxins, influence of drugs taken by mother during pregnancy Environmental factors • Schooling • Interactions with parents / level of intellectual stimulation • Diet/nutrition	0–3
Twin studies	3
Mark allocation: 1 mark for description of how the twin studies are carried out 1 mark for results that have been found 1 mark for conclusions that have been drawn from the results • Description: Show biological/genetic influence by comparing the correlation between IQ scores of identical (monozygotic) twins who share 100% of their DNA with fraternal (dizygotic) twins who share 50% of their DNA. • Results: the IQ correlation of identical twins is higher than fraternal twins. • Conclusion: Intelligence is influenced by biological factors (genetics)	0–3

Adoption studies	3
Mark allocation: 1 mark for description of how the adoption studies are carried out 1 mark for results that have been found 1 mark for conclusions that have been drawn from the results • Description: Show environmental influence by comparing the correlation between IQ scores of twins or siblings who are reared together with twins or siblings who are reared apart. • Results: the IQ correlation of twins/siblings reared together is higher than twins/siblings reared apart. • Conclusion: Intelligence is influenced by environmental factors (schooling, support, etc.) Note: Adoption studies can also look at biological parent and biological child living together, biological parent and biological child living apart, adoptive parent and child living together	0–3
Language is innate	7
Mark allocation: 1 mark for correct theorist 1 mark for correct name of theory Up to 2 marks for description of the theory Up to 3 marks for description of evidence related to the theory Theorist – Chomsky (1 mark) Theory name - Language acquisition device (LAD) (1 mark) Description of theory - • Children are innately predisposed to learn how to talk and understand language (1 mark) • Children are born with a universal grammar – all grammatical information required to produce language (1 mark) Evidence for theory • Language development is innate because: ○ Children master language skill faster than other abilities (1 mark) ○ Children do not simply copy language or learn all phrases, but uncover rules and then create new unlimited phrases (1 mark) ○ Deep structures (universal grammar) is the same in all languages – therefore is genetically determined (1 mark)	0–7
Language is learned	7
Mark allocation: 1 mark for correct theorist 1 mark for correct name of theory Up to 2 marks for description of theory Up to 3 marks for description of evidence related to theory Theorist: Bruner (1 mark) Theory - Language acquisition support system (LASS) (1 mark) Description of theory - • Children learn language through verbal interaction with caregiver, words develop meaning through routines (e.g. bath time) (1 mark) • LASS needs LAD – child component (innate predisposition) and the interaction with others that allows language to be acquired (1 mark)	0–7

Evidence for theory	
- Language develops through interaction during routines such as meal times via: - scaffolding –increasing expectation from caregiver as time progresses - reference – gestures together with words. Eventually words replace gestures - requires joint attention	
Communication skills	3
Well-developed sentences and paragraphs; uses appropriate psychological language with accurate grammar, spelling, and punctuation.	3
Clear everyday language with adequate grammar, spelling, sentences and paragraph structure.	2
Poor literacy skills may interfere with the ability of the marker to understand the response (e.g., very poor spelling, grammar, poor sentence and paragraph structure).	1
Response is unable to be understood.	0
Total	26

Question 12**(23 marks)**

Mrs Mikopolous's Year 12 Psychology class has decided to raise money to donate to the local animal shelter. There are 24 students in the class. The class will be divided into groups and each group must develop a plan for a fundraising activity. Mrs Mikopolous will select the best plan for the whole class to carry out.

Discuss what Mrs Mikopolous needs to consider when organising the class into groups so that they are most productive. Your response should refer to (10 marks)

- brainstorming
- cooperation or competition
- examples of psychological evidence to support your points.

Discuss how the groups' productivity might be reduced and describe what could be done to avoid this. Your response should refer to social loafing and deindividuation. Provide examples and justify your responses by referring to evidence from social psychology.

Question 12	Marks
Improving Productivity	2×5 = 10
Mark allocation: 1 mark for describing concept Up to 2 marks example of application Up to 2 marks for evidence from research/theories that is relevant and supports response Brainstorming Description/definition – “Brainstorming is group problem solving in which members are encouraged to be uninhibited in generating ideas” (Fletcher & Garton, 2007) (1 mark) Up to 2 marks for example/s of application. Any correct/reasonable example of application to be accepted. Example: • To make the brainstorming most effective students should work individually or in small groups to develop their ideas and then develop ideas in a group format. Otherwise ideas are not always well developed and may be swayed or excluded in the group, taking turns to share ideas can take up time, people can get distracted in big groups. Up to 2 marks for evidence of research. Any correct example acceptable. Example includes, but not limited to: Diehl and Stroebe (1987) found that people came up with more ideas on their own than when they were in groups. This was because when in groups, people needed to take turns, were distracted, the discussion got off track etc.	1–5

Cooperation Definition/Description – Working together to achieve the group’s goals (1 mark) Up to 2 marks for example/s of application. Any correct/reasonable example of application to be accepted. - Groups need to be organised by giving each group member a role. Groups work well if they have a group identity. (in-group) (a name, colour scheme, badge, uniform, etc.) Up to 2 marks for evidence of research. Any correct example acceptable. Example includes, but not limited to: - Lewin, Lippit, and White (1939) – effect of group leader on productivity Sherif (1954) Robber’s Cave – build cooperation and cohesion by creating a group identity OR Competition Definition/description – Competition between groups increases productivity whereas competition within groups decreases productivity (1 mark) Up to 2 marks for example/s of application. Any correct/reasonable example of application to be accepted. - Give students a limited time to come up with the most ideas. Offer a reward to the winning group. Up to 2 marks for evidence of research. Any correct example acceptable. Example includes, but not limited to: - Sherif (1954) Robber’s Cave – teams competing for limited resources (realistic conflict theory)	1–5
Reduction of productivity and solutions	2×5=10
Mark allocation: 1 mark for describing concept Up to 2 marks example of application Up to 2 marks for evidence from research/theories that is relevant and supports response Social loafing Description/definition – Tendency for individuals to put less effort in when they are a part of a group (1 mark) Up to 2 marks for example/s of application. Any correct/reasonable example of application to be accepted. - Encourage individual contribution, make individuals accountable by providing individual responsibilities, keep group size small, monitor performance. Give students individual goals if they are from an individualistic culture Up to 2 marks for evidence of research. Any correct example acceptable. Example includes, but not limited to: - Ringelmann 1913- rope pulling experiment – more people pulling the rope meant less individual effort was exerted. - Early 1989 – cultural differences. People from collectivist cultures perform better	1–5

on tasks where they need to achieve a group goal than people from individualistic cultures.	
Deindividuation Description/definition – Usual restraints are abandoned to join in group behaviour, Individuals can show aroused behaviour in groups that they would not show individually (1 mark) Up to 2 marks for example/s of application. Any correct/reasonable example of application to be accepted. - Ensure behaviour expectations are set prior to groups working, have groups develop behaviour norms, make individuals accountable by assigning individual roles that must be fulfilled Up to 2 marks for evidence of research. Any correct example acceptable. Example includes, but not limited to: - Zimbaro (1970) – anonymity allows individuals to take on group behaviour. Individuals who had their identity concealed (anonymous) administered higher electric shocks to other students. - Reicher (1987) – deindividuation increased awareness of group norms seen in a study by Johnson and Downing (1979) – participants were dressed in either nurses uniforms or KKK uniforms. Those dressed in nurses uniforms decreased shock level while participants in the KKK uniforms increased the shock level (these trends were even greater when individual faces were concealed).	1–5
Communication skills	3
Well-developed sentences and paragraphs; uses appropriate psychological language with accurate grammar, spelling, and punctuation.	3
Clear everyday language with adequate grammar, spelling, sentences and paragraph structure.	2
Poor literacy skills may interfere with the ability of the marker to understand the response (e.g., very poor spelling, grammar, poor sentence and paragraph structure).	1
Response is unable to be understood.	0
Total	23

ACKNOWLEDGEMENTS

Section Two

- Question 3** Adapted from: Qasar. [Jarosz, Q.]. (2009, August 11). Neuron hand tuned [Image]. *Wikipedia*. Retrieved from http://en.wikipedia.org/wiki/File:Neuron_Hand-tuned.svg
Used under the Creative Commons Attribution-Share Alike 3.0 Unported license.
- Question 5 (b)** Adapted from: Conservation of liquid [Image]. (n.d.). In Maths to share – CPD for your school. *NCETM Primary Magazine*, (20). Retrieved from www.ncetm.org.uk/resources/22757
- Question 6(a)** Definition of personality from: Fletcher, J., & Garton, A. (2007). *Psychology – Self, others, and society*. Sydney: Pearson Education Australia, p. 497.
- Question 7** Definitions of attribution from: Fletcher, J., & Garton, A. (2007). *Psychology – Self, others, and society*. Sydney: Pearson Education Australia, pp. 54, 442, glossary.

Section Three

- Question 12** Definition of brainstorming from: Fletcher, J., & Garton, A. (2007). *Psychology – Self, others, and society*. Sydney: Pearson Education Australia.

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